Department of Biology
University of Central Florida,
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JORDAN DOWELL

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EDUCATION

PhD University of Central Florida, Integrative Biology 2017-Present
Dissertation: Running Title: "Volatile metabolomic dynamics of the annual clade of *Helianthus*"
Advisors: Chase Mason (chair), Pedro Quintana-Ascencio (vice-chair)

MS University of Nevada, Las Vegas, Ecology & Evolutionary Biology 2015-2019
Thesis: "Landscape scale: inter- and intraspecific variation in plant interactions along a stress gradient in the sheep mountain range of Nevada"
Advisor: Dale Devitt

BS University of Nevada, Las Vegas, Ecology & Evolutionary Biology 2008-2014

RESEARCH EXPERIENCE

Running title: "Volatile metabolomic dynamics of the annual clade of Helianthus",
University of Central Florida 2017-Present
Advisor: Chase Mason

Undergraduate Honors Student Committee Service

- Ana Robinson, "Role of polyploidy in leaf functional trait and secondary metabolite evolution across *Helianthus*", 2019-2020
- Isaac Valdiviezo, "Comparing Leaf Reflectance Analysis Prediction Models Based on Dried Whole Leaf Tissue against Dried Ground Tissue", 2019-2020
- Mari Irving, "Developmental Effects of Terpenes on *Vanessa cardui* at Varying Temperatures", 2019-2020
- Katie Stahlhut, "Genetic control of mycorrhizal colonization in *Helianthus annuus*", 2019-2020
- Danielle De La Pascua, "Reproductive defense trade-offs in the fruits and flowers of the genus *Cornus*", 2018-2019
- Rayner Seavey, "The Hyperaccumulation of Zinc in Sunflowers and its Effect on Disease Resistance", 2018-2019

- Rebekah Davis, "Capillary Electrophoresis buffer optimization for plant tissue analysis", 2018-2019
- Jessica Jowais, "Comparison of secondary metabolite response to *Vanessa cardui* across the genus Helianthus", 2017-2018

Research Associate, University of Nevada, Reno-Cooperative Extension 2017
Advisor: Tammara Wynne
Development of outreach focused experiments concerning domesticated *Solanum lycopersicum* production in the Mojave Desert.

Research Associate, University of Nevada, Las Vegas 2017
Advisors: Lorenzo Apodaca and Dale Devitt
Development and implementation of image analysis-based methods of xylem flow dynamics in urban horticulture trees. Implementation of multidimensional kriging of climate data concerning the ambient effects of photovoltaic power plants on native shrublands.

Restoration Ecology Intern, Great Basin institute, Nevada, Las Vegas 2015
Advisor: Russell Lee Nasrallah
Improvement of highly visible and ecologically important state and national park resources by controlling exotic plants, maintaining hiking trails, and providing an educational resource for park visitors. (Great Basin National Park, Lake Mead National Recreation Area, Spring Mountain Ranch State Park, Desert National Wildlife Refuge, and Pahranagat National Wildlife Refuge).

Undergraduate Research Assistant, University of Nevada, Las Vegas 2012-2013
Advisors: Tereza Jezkova and Javier Rodriguez
Elucidation of phylogeographic and population structure in Mona and Virgin Island Boas, *Chilabothrus monensis* (*Epicrates monenesis)*.
https://doi.org/10.1016/j.ympev.2015.03.019.

TEACHING EXPERIENCE

University of Central Florida Fall 2020
Teaching assistant, Plant Genomics & Biochemistry, Department of Biology
- Advisor: Eric Goolsby
- A mixed-mode, research intensive, joint graduate and undergraduate course. Assorted topics include metabolomics and genomics principles and methods. This course included a student-driven (22 students) research project assessing codon bias, nucleotide diversity and other sequence metrics in relationship to major biosynthetic pathways among cultivated *Helianthus annuus* varieties.

University of Central Florida 2019-2020
Curriculum Development assistant, Department of Biology
- Advisor: Eric Goolsby

- Plant Genomics & Biochemistry, development of instructional materials and laboratories for a new joint graduate and undergraduate course. Assorted topics include chromatography (liquid, gas, & capillary electrophoresis), mass spectrometry, untargeted & targeted metabolomics, metabolic pathway modeling, and machine learning in metabolomics & genomics.

University of Central Florida Fall 2019
UCF Preparing Tomorrow's Faculty program, Karen L. Smith Faculty Center for Teaching and Learning

- Advisor: Masha Krsmanovic
- A professional development workshop dedicated to designing effective college-level courses with inclusive syllabi, active learning focused projects, and proper assessment tools. I designed and develop two full courses focused on chemical ecology and plant biochemistry

University of Nevada, Las Vegas 2015-2017
Graduate Teaching assistant, Department of Biology

- Evolution, a senior level undergraduate course averaging 100 students per semester, focusing on introducing evolutionary biology and the processes that have been (and are currently) responsible for the generation and maintenance of biological diversity.
- Principles of Modern Biology II Lab, an undergraduate laboratory course averaging 60 students per semester, covering the following topics: organismal biology, ecological/evolutionary patterns, and processes.

Bodies: The Exhibition, Las Vegas, NV 2012-2015
Educational Director

- Development of instructional materials for docents and educational outreach materials for assorted topics, including anatomy, physiology, and new developments in the field of medicine and comparative anatomy. During my tenure museum average foot traffic was ~300-600 people per day with a docent staff of 15 individuals.

PUBLICATIONS

Journal Publications
(*undergraduate authors)

- **Dowell, J.A.**, *Clark, E.J., *Pliakas, T.P., Mandel J.R., Burke, J.M., Donovan, L.A., and Mason, C.M., 2019, "Genome wide association mapping of floral traits in cultivated sunflower (*Helianthus annuus*)," *Journal of Heredity*, 110:3 275-286

- **Dowell, J. A.** and Mason, C. M., 2020, "Correlation in plant volatile metabolites: physiochemical properties as a proxy for enzymatic pathways as an alternative biosynthetically informed metric," *Chemoecology*. https://doi.org/10.1007/s00049-020-00322-4

- *De La Pascua, D. R., *Smith-Winterscheidt, C., **Dowell, J. A.**, Goolsby, E. W., and Mason, C. M., 2020, "Evolutionary trade-offs in the chemical defense of floral and fruit tissues across genus *Cornus*," *American Journal of Botany* 107(9): 1260– 1273.

(*in-prep*)

- **Dowell J**, Goolsby E, Temme A, Burke J, Donovan L, Mason M., "Historic breeding practices contribute to variation in leaf specialized metabolism and not ecophysiology in cultivated sunflower (*Helianthus annuus*)".

- Ridenbaugh, R, **Dowell J**, Goolsby E, Sharanowski., "The effects of plant phytochemistry on parasitoid (Hymenoptera: Braconidae) niche breadth".

- *Irving M, *Lim-Hing S, *Stanford H, *Urrea M, **Dowell J**, Goolsby E, Mason C. "Developmental effects of terpenes on *Vanessa cardui* at varying temperatures."

- *Stahlhut K, Temme A, **Dowell J**, Burke J, Goolsby E, Mason C. "Genetic architecture of arbuscular mycorrhizal colonization in *Helianthus annuus*."

- *Valdiviezo M, **Dowell J**, Goolsby E, Mason C. "Validation and optimization of hyperspectral reflectance analysis-based predictive models for the determination of plant functional traits in *Cornus*, *Rhododendron*, and *Salix*."

- *Robinson A, Majumder S, **Dowell J**, Goolsby E, Mason C. "The role of polyploidy in secondary metabolite evolution across wild *Helianthus*"

- **Dowell, J.A.**, Devitt, D., "Landscape scale: inter-and intraspecific variation in plant interactions along stress gradients".

Conference Oral/Poster Presentations

- **Dowell, J. A.**, "Evolution & diversification of plant-plant communication: An intermediate hypothesis", Plant Biology 2020

- **Dowell, J. A.**, Mason, C. M. "Correlation in plant volatile metabolites: physiochemical properties as a proxy for enzymatic pathways and an alternative metric of biosynthetic constraint", Botany 2020, ABSTRACT ID-576

- *De La Pascua, D. R., *Smith-Winterscheidt, C., **Dowell, J. A.,** Goolsby, E. W., Mason, C. M.,S, "Evolutionary trade-offs in the chemical defense of floral and fruit tissues across genus *Cornus*", Botany 2020, ABSTRACT ID-166

- *Stahlhut, K. **Dowell, J. A.,** Temme, A., Burke, J., Goolsby, E., Mason, C. M., "Genetic control of arbuscular mycorrhizal colonization in *Helianthus annuus*", Botany 2020 ABSTRACT ID-137

- *Stahlhut, K. **Dowell, J. A.,** Mason, C. M., "Genetic control of mycorrhizal colonization in *Helianthus annuus*", Botany 2019, ABSTRACT ID-469

- *Davis, R., **Dowell, J. A.,** and Mason, C. M., "Describing Photosynthetic Diversity in Cultivated Sunflower", Botany 2019, ABSTRACT ID-385

- *Valdiviezo, M. I., **Dowell, J. A.,** and Mason, C. M., "Comparing Leaf Reflectance Analysis Prediction Models Based on Dried Whole Leaf Tissue against Dried Ground Tissue", Botany 2019, ABSTRACT ID-766.

- **Dowell, J. A.** and Mason, C. M., "An evolutionarily relevant definition of 'Eavesdropping' and 'Communication", International society of chemical ecology, 2019

- **Dowell, J. A.** and Mason, C. M., "Impacts of physical chemistry on biosynthetic constraints of plant volatile profiles", International society of chemical ecology, 2019

- **Dowell, J.A.,** *Clark, E.J., *Pliakas, T.P., Mandel J.R., Burke, J.M., Donovan, L.A., and Mason, C.M., "Genome wide association mapping of floral traits in cultivated sunflower (*Helianthus annuus*)," Botany, 2018, ABSTRACT ID-295.

WORKSHOPS AND INVITED LECTURES

- **Lecture,** "Quantitative genetics in cultivated sunflower (*Helianthus annuus*).", University of Central Florida, Department of Biology: Evolutionary Biology course, October 2020

- **Lecture,** "Evolution & diversification of plant-plant communication: An intermediate hypothesis", Plant Biology 2020, MAC Symposium 3: Evo-Devo 2020: Case Studies in Diversity. July 2020

- **Lecture**, "The language of life: chemically mediated interactions in plant ecology & evolution", Niagara University, Early career researcher diversity seminar series, November 2019.

- **Workshop**, "Comparative plant metabolomics & Bayesian hierarchal clustering analysis", University of Central Florida, Department of Biology, November 2019.

- **Lecture**, "Sassy sages and gossiping goldenrods: recent advances in plant volatile communication", Florida Native Plant Society, Florida Native Plant Month, October 2019.

- **Lecture**, "Volatile metabolomics of the annual clade of *Helianthus*", University of Central Florida, Department of Biology, September 2019.

- **Workshop**, "Comparative analytical techniques in Plant Metabolomics", University of Central Florida, Department of Biology, June 2019.

- **Lecture**, "Sassy sages and gossiping goldenrods: recent advances in plant volatile communication", Florida Native Plant Society, Tarflower Chapter, May 2019.

- **Workshop**, "Introduction to Random Forest models", University of Central Florida's Biology Graduate Student Association data science seminar, February 2018.

- **Lecture**, "Landscape Scale: inter-and intraspecific variation in plant interactions along a stress gradient in the sheep mountain range", University of Nevada, Las Vegas, Graduate Student Seminar series, May 2017.

GRANTS, HONORS AND AWARDS

UCF College of Science Visiting Scholars program 2020-Reocurring
- $32,000 per academic year, Internal funding allocation (**funded**)
- Co-author Ian Will
- Initiative to supply funding to bring in historically excluded scholars to give research seminars and supply a mentorship opportunity for historically excluded undergraduate and graduate students.

Florida Education Fund Dissertation Fellowship 2020-2021
- $12,000 External Fellowship (**funded**)

University of Washington's Statistical Genetics Summer Institute Scholarship 2020
- $900 External Workshop Scholarship (**funded**)

University of Central Florida's Doctoral Research Support Award 2019
- $5,000 Internal Research Grant (unfunded)
- "Volatile metabolomic dynamics of the annual clade of *Helianthus*"

Bill and Melinda Gates Foundation 2019
- $1,000 External travel award **(funded)**
- Institute for Teaching and Mentoring: Compact for Faculty Diversity

U.S.D.A, Sustainable Agriculture Research and Education 2019
- $16,499 External research grant (unfunded)
- "Genome-wide association of the volatile metabolome of cultivated *Helianthus annuus*"

Society for the Study of Evolution, RC Lewontin Early Award 2019
- $2,499 External research grant (unfunded)
- "Evolutionary dynamics of biotic-induced responses in the annual clade of *Helianthus*"

University of Central Florida, Biology Graduate Student Association 2019

- $300 Internal travel award **(funded)**

University of Central Florida, Biology Department 2019

- $250 Internal travel award **(funded)**

Bill and Melinda Gates Foundation Millennium Graduate Fellow 2015-2020

- $175,000 External Fellowship **(funded)**

University of Central Florida's Doctoral Conference Support 2017

- $4,300 Internal Grant **(funded)**

University of Nevada, Las Vegas: Graduate and Professional Society 2016

- $1,000 Internal Research Grant **(funded)**

Bill and Melinda Gates Foundation 2016

- $1,000 External travel award **(funded)**
- Institute for Teaching and Mentoring: Compact for Faculty Diversity

Bill and Melinda Gates Foundation Millennium Undergraduate 2009-2014

- $150,000 External Scholarship **(funded)**

Silver State Millennium Foundation Scholarship 2009-2011

- $2,500 External Scholarship **(funded)**

PROFESSIONAL AFFILIATIONS

University of Nevada, Las Vegas, 2017-2019
Research Associate Provided statistical consultation for evaluation of landscape-scale spatial dependencies of vegetation dynamics.

University of Nevada, Reno-Cooperative Extension, 2017-2018
Research Associate Provided statistical consultation for design of outreach orientated desert agricultural projects.

PROFESSIONAL SERVICE

- **American Society of Plant Biology panel member; "Interested in Graduate School"**
  - July 2020
  - Answered questions concerning graduate school funding and career options for an audience of ~100 undergraduate participants.
- **Science Pawdcast guest speaker; working title: "Talking plants & challenges of BIPOC in graduate education"**
  - July 2020

    - Currently in production
    - Episode XX link:
- **Reviewing Editor for *Applications in Plant Sciences* 2020-Present**
    - Responsibilities include: Identifying suitable reviewers, synthesizing external reviews, working alongside associate editors on decision processes, proposing topics for review articles and special issues, and providing input on new journal activities.
- **Consultant**
    - SEE Turtles organization BIPOC scholarship fund 2020-Reocurring
        - $10,000
        - Supplied guidance on barriers for BIPOC engaging in field study summer programs and developed a funding schema to house and pay students.
- **Reviewer**
    - Oikos, Journal of Chemical Ecology, Chemoecology
- **Biology Graduate Student Association**
    - President, University of Central Florida, Aug. 2019-present
    - Secretary, University of Central Florida, Aug. 2018-2019

- **UCF's Biology Integrated Orlando Training and Enrichment Camp**
    - Plant Science coordinator, Jun. 2018
    - Developed and taught a weeklong curriculum of plant science focused experiments to engage over 30 high school students from the surrounding Orlando metropolitan area in current research techniques, such as genomics and metabolomics.
- **Botanical Society of America Graduate School Career Panel (UCF Chapter)**
    - September 2019
    - Answered questions concerning graduate school funding and career options for an audience of 50 undergraduate participants.

- **Plants Beyond Limits Conference**
    - Graduate Student Coordinator, University of Central Florida, Aug. 2017
        - Initiated, organized, and funded the first student-led conference of Plants Beyond Limits at UCF with ~500 attendees, 20 speakers, and 15 submitted posters from graduate students and postdocs

REFERENCES

- **Chase Mason, Ph. D.**
    - **Assistant Professor in the Department of Biology at the University of Central Florida**
        - Email
            - chase.mason@ucf.edu
        - Phone number
            - (407)-801-3823
        - Relationship
            - Dr Mason is my major advisor for my doctoral studies.

- **Eric Goolsby, Ph. D.**
    - **Assistant Professor in the Department of Biology at the University of Central Florida**
        - Email
            - Eric.goolsby@ucf.edu
        - Phone number
            - (478)-960-5020
        - Relationship
            - Dr. Goolsby is a member of my doctoral committee, with whom I have several ongoing collaborations both in research and education.
- **Laurence Von Kalm, Ph. D.**
    - **Associate Professor & Associate Chair in the Department of Biology at the University of Central Florida**
        - Email:
            - Laurence.vonkalm@ucf.edu
        - Phone number
            - (407)-823-6684
        - Relationship
            - During my tenure as president and secretary of the University of Central Florida's Biology Graduate Student Association I have worked hand in hand with Dr. Von Kalm on issues concerning graduate student wellbeing, education, and departmental policy.